Cartoon Eyes Emotion Predictor

Tiffany Ho
The University of Texas at Dallas
tiffany.ho@utdallas.edu

Abstract

Many studies have been done on human facial recognition and its variations. However, there is not much information on predicting the emotions of cartoon characters based solely on their eyes. This application based project delves into this topic. The proposed idea is to research animated facial expression recognition, then proceed with determining a method to implement detecting the emotions of cartoon characters using only their eyes. This project will focus on three emotions: happy, sad, and angry, using images of cartoon characters found online that will then be cropped to show only their eye region. The method utilized will be HOG (histogram of oriented gradients) feature extraction with an SVM (support vector machine) classifier. Specifically, image features will be extracted with HOG feature extraction, capturing the edges and shapes of the eyes, which will then be fed to the SVM for classification. While this method was only able to achieve 55.56% accuracy, further improvements such as a larger and more diverse training set may be able to significantly increase performance.

1. Introduction

With the advancements in facial recognition and artificial intelligence in the past decade, AI generated content has become increasingly popular. While facial expression recognition technology in humans has made many advancements, animated expression evaluation has remained relatively less studied. By delving into predicting the emotions of cartoon characters through only their eye region, further improvements can be made in the sector of AI generated content recognition through overlapping similarities of the two topics.

In this project, I will be predicting the emotions of cartoon characters through their eyes. After research on similar topics, I have decided to use HOG feature extraction along with an SVM classifier. Specifically, HOG feature extraction will capture the edges and shapes of the eyes, which will then be fed to the SVM for classification.

2. Related Work

In this project, I mainly referenced two related works as well as experience from my own previous similar project from CS 4391:

1. Deep Learning for Emotion Recognition in Cartoons by John Wesley Hill (dissertation) [2]

2. Facial Expression Recognition of Animated Human Characters by Guoxu Liu, Hui Jin, Hailong Jiang, and Jae Ho Kim (ACM 2020) [3]

3. Animated Facial Expression Recognition by Tiffany Ho (my own previous project from CS 4391: Computer Vision - undergraduate)

The first paper, Deep Learning for Emotion Recognition in Cartoons by John Wesley Hill, is a dissertation with the goal being to examine if cartoon emotions can be detected in the same way that human faces can.

The paper details facial emotion recognition when combined with classifiers and deep neural networks, the neural network being CNN (convolutional neural network). This method was found to achieve a validation rate as high as 59% and a recognition rate of 56%. The system implemented in this paper focused on classifying three emotions: happy, angry, and surprised, with an accuracy of 80% for the three emotions.

The network was found to be shallower than expected and had some overfitting issues when using more layers of convolution and max pooling. While increasing the dropout helped slightly, the author found that including a wider variety of emotions as well as a larger dataset in general would yield an immense improvement in performance.

The second paper, Facial Expression Recognition of Animated Human Characters by Guoxu Liu, Hailong Jiang, and Jae Ho Kim, proposed an improved algorithm for recognizing the facial expression of animated human characters based on LBP (local binary pattern) and SVM (support vector machine). This method was tested on animated human characters from popular animations, as this paper has

a main focus on the movie animation industry. The results of this new method showed a significant improvement in performance compared to other known methods.

The study included the following six emotions: anger, disgust, fear, happy, sad, and surprise. To remedy the shortcomings of standard a LBP descriptor on non-uniform texture, the study proposed multiple-range LBP and multiple region LBP, and an SVM was used for classification. While traditional LBP only calculates intensity variations on neighbors in a 3x3 window, multiple radius LBP extends the neighborhood for LBP with various radius sizes, as shown below in Figure 2 from the paper.

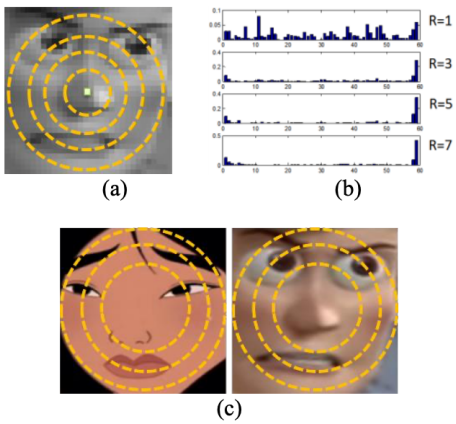


Figure 2. Facial feature calculation with the proposed multiple radius LBP and the corresponding histograms.

In addition to multiple radius LBP, the authors also proposed multiple overlapped region LBP, which analyzes various overlapping regions of the face, as shown below in Figure 3 from the paper.

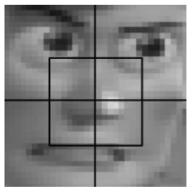


Figure 3. Multiple regions for obtaining facial components feature by LBP.

The new method proposed by the authors was found to be significantly more accurate compared to existing methods due to increased image evaluation area.

My last referenced work was my own previous project from CS 4391, Animated Facial Expression Recognition. This previous project was similar to my current one, except it evaluated the entire face of animated two-dimensional cartoon characters using a CNN. It also evaluated three emotions: happy, angry, and sad, with 20 training images for each emotion. This project was only able to each 50% accuracy, as seen in the image below from my previous project results.

From this idea, I decided to modify the method and only evaluate the eye portion for my current project. While the current project only achieved slightly higher accuracy at 55.56%, I would consider it meaningful, as only having the eye region for predictions makes evaluation more difficult, especially in animated characters.

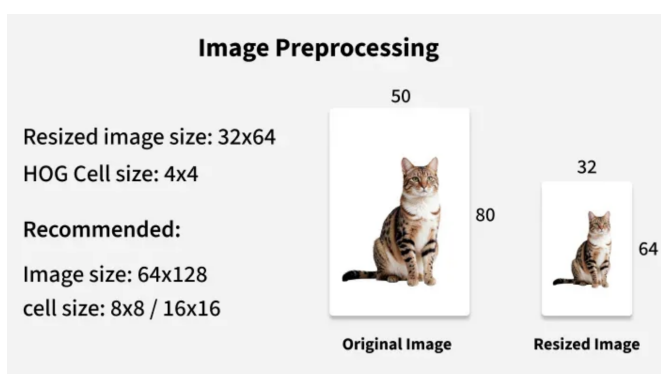
3. Method

For this project, I utilized HOG feature extraction in combination with an SVM classifier. The feature vectors from HOG are used as input for the SVM classifier.

3.1. HOG

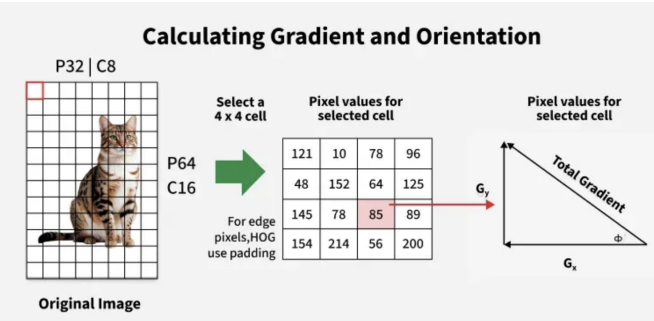
HOG (histogram of oriented gradients) captures an images edges and shapes and converts them into numerical feature vectors. The following images are from GeeksforGeeks' site [1].

The images are first converted to grayscale to reduce complexity and resized to 50x50 pixels to ensure consistency across all input images.

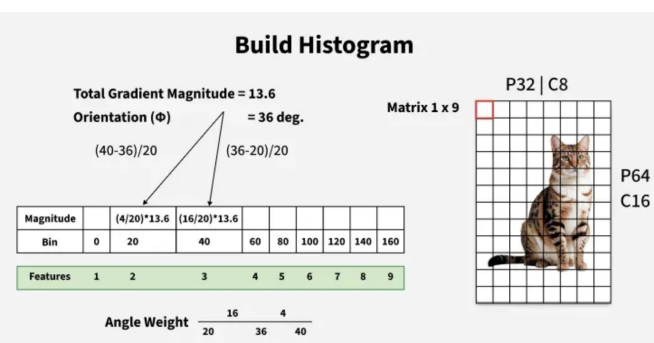


Then, the gradient is computed for each pixel. This is the edge direction + magnitude in both X and Y directions.

- ***Gradient magnitude:*** Magnitude $= \sqrt{G_x^2 + G_y^2}$
- ***Gradient orientation:*** Orientation $= \arctan2(G_y, G_x)$



Next, orientation histograms are created for each cell in the image. These are histograms of the edge directions previously calculated. For each pixel, the gradient magnitude and orientation are calculated (orientation values range from 0 to 180 degrees). The 0-180 degrees of orientation are divided into 9 histogram bins of 20 degrees each. Each pixel's magnitude is then assigned to the two nearest bins. This is determined by the distance from each orientation to the bin center. This is done for all pixels in the cell. Each cell will now have a 1x9 feature vector representing the distribution of edge directions. These are the feature vectors that will be fed as input to the SVM classifier.



3.2. SVM

The SVM (support vector machine) classifies the emotions using the HOG feature vectors previously computed. This finds the best decision boundary between classes, the emotions being the classes. The goal is to maximize the gap between different emotions for clearer classification.

Before training the model, dataset preparation must be done for the training images. After extracting the HOG features, numeric labels will be assigned to each emotion: happy: 0, sad: 1, angry: 2. X_train will be the feature vectors and y_train will be the numeric labels.

The SVM will then learn a decision boundary separating emotions. The output is a numeric label mapped back to its corresponding emotion. For HOG, a linear SVM is used, which defines a hyperplane. The SVM seeks the boundary that maximizes the margin, which is the maximum distance between the two closest HOG feature vectors of different classes.

3.3. HOG + SVM

This combination of HOG and SVM builds the model that finds the maximum decision boundary (hyperplane) separating the classes (emotions). During testing, the test image will go through HOG feature extraction, then classified. During classification, the HOG feature vector is dropped into the SVM feature space based on the vector values. The model checks which side of the boundary this point falls on and returns a label representing the closest emotion match based on its distance from the decision boundary.

4. Experiments

4.1. Dataset

For my experiment, I focused on three emotions: happy, sad, and angry. My dataset consisted of training and test images.

4.2. Training

For my training set, I collected 20 images for each emotion. These were images of cartoon characters found online that were then cropped to show only their eye region. Characters with non-standard eye structures, glasses, and those with only one eye are also included. The characters consisted of both two-dimensional and three-dimensional characters from popular movies and tv. Some images from my training dataset are shown below.

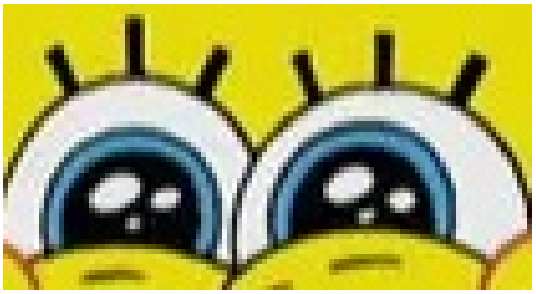


Figure 1. Happy Spongebob

Figure 2. Happy Patrick

Figure 3. Sad Perry

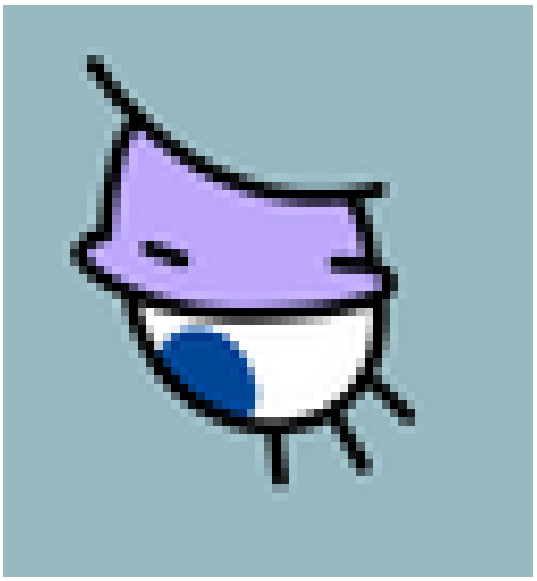


Figure 4. Sad Pearl

Figure 5. Angry Nick

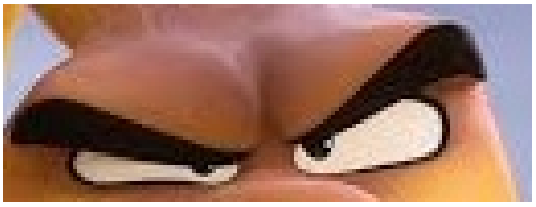


Figure 6. Angry Snake

4.3. Testing

The test set consisted of nine images total, with three images for each emotion. Some images from my test dataset are shown below.

Figure 7. Happy Squidward

Figure 8. Sad Mrs. Puff

Figure 9. Angry Tina Belcher

4.4. Results

Results found that the angry images were the most ambiguous. Specifically, the angry label was confused for happy twice, and a sad image was confused for angry twice. Overall, the model was able to achieve 55.56% accuracy. A confusion matrix with the results was generated, with the rows representing true (actual) labels, and the columns representing the model's predicted labels. The model predictions and the confusion matrix are shown below.

```
Predictions
happy squidward.jpg: Predicted = happy, True = happy
angry lisa.jpg: Predicted = angry, True = angry
angry bob.jpg: Predicted = sad, True = angry
sad chris.jpg: Predicted = sad, True = sad
sad mrs puff.jpg: Predicted = sad, True = sad
angry tina.jpg: Predicted = sad, True = angry
sad bart.jpg: Predicted = sad, True = sad
happy lisa.jpg: Predicted = angry, True = happy
happy mrs puff.jpg: Predicted = angry, True = happy

Model Accuracy Percentage: 55.56%

Confusion Matrix:
       happy  sad  angry
happy      1    0      2
sad        0    3      0
angry      0    2      1
```

Figure 10. Results and Confusion Matrix

Interpretation of confusion matrix:

Of the happy test images, 1 was predicted as happy, 0 as sad, and 2 as angry.

Of the sad test images, 0 were predicted as happy, 3 as sad, and 0 as angry (all predictions were correct).

Of the angry test images, 0 were predicted as happy, 2 as sad, and 1 as angry.

5. Conclusion

In summary, this project utilized HOG feature extraction with an SVM classifier to predict the emotions of cartoon characters based on only their eye region. The emotions evaluated were happy, sad, and angry, and the model was able to achieve 55.56% accuracy.

While the accuracy is not too high, a larger and more diverse training dataset would significantly improve performance. With more variety in images' eye expression, the model will be more exposed to different edges and angles of the eyes from each emotion. Naturally, a larger dataset will provide similar benefits. This will allow the model to better discern which emotion is being expressed. With more resources and time, significant improvements in emotion detection can be made.

References

[1] GeeksforGeeks. Histogram of oriented gradients. https://www.geeksforgeeks.org/computer-vision/histogram-of-oriented-gradients/, 2025. 2

[2] John Wesley Hill. Deep learning for emotion recognition in cartoons, 2017. https://hako.github.io/dissertation/. 1

[3] Guoxu Liu, Hui Jin, Hailong Jiang, and Jae Ho Kim. Facial expression recognition of animated human characters. In *Proceedings of the International Conference on Machine Learning and Cybernetics (ICMLC)*, Shenzhen, China, 2020. ACM. https://dl.acm.org/doi/10.1145/3383972.3384026. 1